The Induced Correlations of Zadoff-Chu Sequences

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***Abstract*—The induced correlations of a pair of sequences are defined as the full-period correlations between the linear phase-shifting sequences of one of the given pair and the other one. This concept was introduced in the analysis of their partial-period correlations which are an important performance measure of communication systems employing them. In this paper, we investigate the induced correlations of Zadoff-Chu sequences in a transform approach. For a pair of Zadoff-Chu sequences, we first compute the spectrums of their induced correlations. By taking the inverse discrete Fourier transform (IDFT) on these spectrums, we then derive their induced correlations in a closed form. As a result, we show that their induced correlations can be viewed as expanded and scaled Zadoff-Chu sequences of a smaller period. Not only does our approach give the magnitudes of the induced correlations of Zadoff-Chu sequences, but it also gives the phase information which is applicable to computation of their partial-period correlations.**

I. Introduction

Pseudo-random sequences with good correlation properties have many applications to communication systems [1]. In these systems, it is required to employ a family of pseudo-random sequences with low correlation. There are two types of correlation: *full-period correlation* and *partial-period correlation*. Full-period correlation has been widely studied so far, but partial-period correlation has not received much attention in spite of its importance in real situations [2]-[3]. In code-division multiple-access (CDMA) systems, each data symbol is divided into B chips. When a sequence of period $N > B$ is employed as a spreading sequence, the actual correlation commonly computed at the receiver is not its full-period correlation, but its partial-period correlation [2]. In such a case, the magnitudes of partial-period correlations are needed to be low. Partial-period correlation also plays an important role in the synchronization of communication systems [4].

Paterson and Lothian [3] linked the partial-period correlations to the discrete Fourier transform (DFT) of the so-called window sequence via a character sum. Recently, Lee *et al.* [5] generalized the Paterson-Lothian approach by introducing the *linear phase-shifting sequences* of one of a given pair of sequences and analyzing their full-period correlations, called the *induced correlations*, with the other one. Their approach is particularly useful in the sense that the partial-period correlations of sequences can be analyzed in terms of their induced correlations.

Zadoff-Chu sequences [6], [7] are one of the best known and most widely used families of polyphase sequences [8]. They have been used as spreading sequences in CDMA systems [9] and as preamble sequences in several communication systems such as long-term evolution (LTE) [10], because they have ideal full-period autocorrelation in addition to their existence for any period N. Furthermore, they have optimal full-period cross-correlation with respect to the Sarwate bound [11] under some specific conditions. Lee *et al.* [5] computed completely the magnitudes of the induced correlations of Zadoff-Chu sequences and employed them to derive an upper bound on the magnitudes of the partial-period correlations of Zadoff-Chu sequences.

In this paper, we investigate the induced correlations of Zadoff-Chu sequences in a transform approach. Given a pair of Zadoff-Chu sequences, we first compute the spectrums of their induced correlations. By taking the inverse discrete Fourier transform (IDFT) on these spectrums, we then derive their induced correlations in a closed form. As a result, we show that their induced correlations can be viewed as expanded and scaled Zadoff-Chu sequences of a smaller period.

The outline of the paper is as follows. Section II presents some preliminaries to induced correlations. In Section III, we review Zadoff-Chu sequences and their corresponding spectrums, and then investigate the induced correlations of Zadoff-Chu sequences in the transform domain. We then present their induced correlations in a closed form in Section IV. Finally, we give some concluding remarks in Section V.

II. Preliminaries

Throughout the paper, we denote by $\langle x \rangle_y$ the least nonnegative residue of x modulo y for an integer x and a positive integer y, and denote by $\mathbb{Z}_N$ the set of nonnegative residues modulo N for a positive integer N.

Let $\{a(n)\}_{n=0}^{N-1}$ be a complex-valued sequence of period N. It will be simply denoted by $\{a(n)\}$, if there is no confusion. Let $\mathcal{F}(\cdot)$ and $\mathcal{F}^{-1}(\cdot)$ denote the discrete Fourier transform (DFT) and inverse discrete Fourier transform (IDFT) operators, respectively. The DFT sequence $\{\hat{a}(l)\} \triangleq \mathcal{F}(\{a(n)\})$ of $\{a(n)\}$ is defined as

$$\hat{a}(l) = \sum_{n=0}^{N-1} a(n) W_N^{nl}, \quad 0 \le l < N$$

where $W_N = \exp(2\pi\sqrt{-1}/N)$. The sequence $\{\hat{a}(l)\}$ is simply called the *spectrum* of $\{a(n)\}$. Conversely, the sequence $\{a(n)\}$ can be obtained via the IDFT as $\{a(n)\} \triangleq$

$\mathcal{F}^{-1}(\{\hat{a}(l)\})$, that is,

$$a(n) = \frac{1}{N}\sum_{l=0}^{N-1} \hat{a}(l) W_N^{-nl}, \quad 0 \le n < N. \tag{1}$$

Let $\{a(n)\}$ and $\{b(n)\}$ be two complex-valued sequences of period N. The *(full-period) cross-correlation* $\theta_{a,b}(\tau)$ between $\{a(n)\}$ and $\{b(n)\}$ is defined as

$$\theta_{a,b}(\tau) = \sum_{n=0}^{N-1} a(n) b^*(n+\tau)$$

for an integer $0 \le \tau < N$, where $*$ denotes the complex conjugation and all the operations among the indices are computed modulo N. When $a(n) = b(n)$ for all $n \in \mathbb{Z}$, the set of integers, $\theta_{a,b}(\tau)$ is called the *(full-period) autocorrelation* of $\{a(n)\}$ and is simply denoted by $\theta_a(\tau)$.

The partial-period correlation which is computed within a window with a specific size may be considered as a general measure of correlation. For an integer $0 \le \tau < N$, an integer $0 \le k < N$ and an integer $1 \le B \le N$, the *partial-period cross-correlation* $R_{a,b}(\tau, k; B)$ between $\{a(n)\}$ and $\{b(n)\}$ is defined as [1]

$$R_{a,b}(\tau, k; B) = \sum_{n=k}^{k+B-1} a(n) b^*(n+\tau).$$

Here, k and B denote the initial position and the size of the correlation window, respectively. Similarly, if $a(n) = b(n)$ for all $n \in \mathbb{Z}$, $R_{a,b}(\tau, k; B)$ is called the *partial-period autocorrelation* of $\{a(n)\}$, and is simply denoted by $R_a(\tau, k; B)$. Clearly, we have $R_{a,b}(\tau, k; N) = \theta_{a.b}(\tau)$ for any $0 \le \tau < N$ and any $0 \le k < N$.

For two positive integers N and B with $B \le N$ and an integer $0 \le k < N$, the window sequence $\{\lambda_{k,B}(n)\}_{n=0}^{N-1}$ of length N is defined as $\lambda_{k,B}(n) = \mathbf{I}_{A(k,B)}(n)$ where $A(k, B)$ is the subset of $\mathbb{Z}_N$ given by

$$A(k,B) = \begin{cases} \{n \,|\, k \le n \le k+B-1\}, & \\ \qquad\qquad\qquad \text{if } k+B-1 < N & \\ \{n \,|\, 0 \le n \le \langle k+B-1\rangle_N & \\ \qquad \text{or } k \le n < N\}, & \\ \qquad\qquad\qquad \text{if } k+B-1 \ge N & \end{cases}$$

and $\mathbf{I}_S(\cdot)$ denotes the indicator function of S given by

$$\mathbf{I}_S(n) = \begin{cases} 1, & \text{if } n \in S \\ 0, & \text{otherwise.} \end{cases}$$

Paterson and Lothian [3] expressed partial-period correlations in terms of the spectrum $\{\hat{\lambda}_{k,B}(l)\}$ of $\{\lambda_{k,B}(n)\}$. Lee *et al.* [5] generalized their idea and proposed a representation of partial-period correlations. For an integer $0 \le l < N$, we define the *lth linear phase-shifting sequence* $\{a^l(n)\}_{n=0}^{N-1}$ of a sequence $\{a(n)\}$ as

$$a^l(n) \triangleq a(n) W_N^{-nl}, \quad 0 \le n \le N-1.$$

Then the partial-period correlation $R_{a,b}(\tau, k; B)$ between $\{a(n)\}$ and $\{b(n)\}$ can be represented as

$$R_{a,b}(\tau, k; B) = \frac{1}{N}\sum_{l=0}^{N-1} \hat{\lambda}_{k,B}(l)\, \theta_{a^l,b}(\tau). \tag{2}$$

Hence, it is possible to compute $R_{a,b}(\tau, k; B)$ if we know the spectrum of the window sequence $\{\lambda_{k,B}(n)\}$ and the full-period correlations at displacement τ between the lth linear phase-shift sequence $\{a^l(n)\}$ and $\{b(n)\}$ for $0 \le l < N$. Since $\{\hat{\lambda}_{k,B}(l)\}$ has nothing to do with both $\{a(n)\}$ and $\{b(n)\}$, the generic properties of their partial-period correlations are reflected in the full-period correlations $\{\theta_{a^l,b}(\tau)\}_{\tau=0}^{N-1}$ for $0 \le l < N$. In particular, $\{\theta_{a^l,b}(\tau)\}_{\tau=0}^{N-1}$ is referred to as the *lth induced correlation* of $\{a(n)\}$ and $\{b(n)\}$.

III. Spectrums of the Induced Correlations of Zadoff-Chu Sequences

As shown in the previous section, the induced correlations of $\{a(n)\}$ and $\{b(n)\}$ play a key role in analyzing their partial-period correlations. In this section we will investigate the induced correlations $\{\theta_{a^l,b}(\tau)\}$ in the transform domain when both $\{a(n)\}$ and $\{b(n)\}$ are Zadoff-Chu sequences of the same period.

Definition 1 ([6], [7]). *For a positive integer N and an integer $1 \le r < N$ with $\gcd(N, r) = 1$, the Zadoff-Chu sequence $\{c_r(n)\}$ of period N is defined as*

$$c_r(n) = W_N^{\frac{rn(n+\langle N\rangle_2)}{2}}.$$

Lee *et al.* [5] completely computed the magnitudes of the induced correlations $\{\theta_{c_r^l,c_s}(\tau)\}_{\tau=0}^{N-1}$, $0 \le l < N$, of two Zadoff-Chu sequences $\{c_r(n)\}$ and $\{c_s(n)\}$, although they discarded the phase information on their induced correlations due to the difficulty of computation. In general, the phase information on the induced correlations of sequences may also be useful for computing their partial-period correlations in (2). Hence, it is interesting to evaluate the induced correlations of Zadoff-Chu sequences without loss of their phase information. To do so, we will first find the spectrums of their induced correlations in the transform domain. By taking the IDFT on their spectrums, we then derive the induced correlations $\{\theta_{c_r^l,c_s}(\tau)\}$. Throughout this section, we denote by m the transform domain index corresponding to τ since l has already been used as the index for the lth linear phase-shifting sequence. Thus, for $0 \le l < N$, we have

$$\mathcal{F}(\{\theta_{c_r^l,c_s}(\tau)\}_{\tau=0}^{N-1}) = \{\hat{\theta}_{a^l,b}(m)\}_{m=0}^{N-1}.$$

Definition 2 ([12]). *Let n be an odd positive integer with factorization $n = p_1^{e_1} p_2^{e_2} \cdots p_K^{e_K}$, where p_i, $i = 1, 2, \ldots, K$, are pairwise distinct primes. For an integer a, the Jacobi symbol $\left(\frac{a}{n}\right)_J$ is defined as*

$$\left(\frac{a}{n}\right)_J = \left(\frac{a}{p_1}\right)^{e_1} \left(\frac{a}{p_2}\right)^{e_2} \cdots \left(\frac{a}{p_K}\right)^{e_K}$$

where $\left(\frac{a}{p}\right)$ *represents the Legendre symbol, defined for all integers* a *and all odd primes* p *by*

$$\left(\frac{a}{p}\right) = \begin{cases} 0, & if\ \langle a\rangle_p = 0 \\ 1, & if\ \langle a\rangle_p \neq 0\ and\ a = x^2\ mod\ p \\ & \quad for\ some\ integer\ x \\ -1, & otherwise. \end{cases}$$

Conventionally, we set $\left(\frac{a}{1}\right)_J = 1$ for any integer a, although 1 is not a prime. The Jacobi symbol can be efficiently calculated in an iterative way without actual calculation of the corresponding Legendre symbols [13].

Definition 3 ([14], [15]). *For a positive integer L, the L-fold expander* $E_L(\cdot)$ *converts a sequence* $\{a(n)\}_{n=0}^{N-1}$ *of period* N *into the sequence* $\{a_{E_L}(n)\}_{n=0}^{LN-1} \triangleq E_L\left(\{a(n)\}_{n=0}^{N-1}\right)$ *of period* LN*, defined by*

$$a_{E_L}(n) = \begin{cases} a(n/L), & if\ \langle n\rangle_L = 0 \\ 0, & otherwise \end{cases} \tag{3}$$

for $0 \leq n < LN$ *where* L *is referred to as the expanding rate.*

Definition 4. *For a positive integer* M*, the* M*-fold repeater* $R_M(\cdot)$ *converts a sequence* $\{a(n)\}_{n=0}^{N-1}$ *of period* N *into the sequence* $\{a_{R_M}(n)\}_{n=0}^{MN-1} \triangleq R_M\left(\{a(n)\}_{n=0}^{N-1}\right)$ *of period* MN*, given by*

$$a_{R_M}(n) = a(\langle n\rangle_N), \quad 0 \leq n < MN$$

where M *is referred to as the repeating rate.*

The following lemma lists some well-known properties in the literature.

Lemma 5 ([8], [15]). *Let* $\{a(n)\}$ *and* $\{b(n)\}$ *be two complex-valued sequences of period* N*.*

a) *Their full-period cross-correlation* $\{\theta_{a,b}(\tau)\}_{\tau=0}^{N-1}$ *is given by*

$$\{\theta_{a,b}(\tau)\} = \{a(-\tau)\} \circledast \{b^*(\tau)\}$$

where $\circledast$ *stands for the circular convolution defined for* $\{z(n)\} = \{x(n)\} \circledast \{y(n)\}$ *as*

$$z(n) = \sum_{i=0}^{N-1} x(i)y(\langle n-i\rangle_N), \quad 0 \leq n < N.$$

b) *The spectrum* $\{\hat{\theta}_{a,b}(m)\}_{m=0}^{N-1}$ *of* $\{\theta_{a,b}(\tau)\}_{\tau=0}^{N-1}$ *is given by*

$$\hat{\theta}_{a,b}(m) = \hat{a}(-m)\hat{b}^*(-m), \ 0 \leq m < N.$$

c) *For any integers* n_0 *and* l_0*, we have*

$$\mathcal{F}(\{a(n+n_0)\}) = \{\hat{a}(l)W_N^{-n_0 l}\}$$

and

$$\mathcal{F}(\{a(n)W_N^{l_0 n}\}) = \{\hat{a}(l+l_0)\}.$$

d) *For any positive integers* L *and* M*, we get*

$$\mathcal{F}\left(E_L(\{a(n)\}_{n=0}^{N-1})\right) = R_L\left(\{\hat{a}(l)\}_{l=0}^{N-1}\right)$$

and

$$\mathcal{F}\left(R_M(\{a(n)\}_{n=0}^{N-1})\right) = M \cdot E_M\left(\{\hat{a}(l)\}_{l=0}^{N-1}\right).$$

Given two Zadoff-Chu sequences $\{c_r(\tau)\}$ and $\{c_s(\tau)\}$ of period N, the spectrum of their lth induced correlation $\{\hat{\theta}_{c_r^l,c_s}(m)\}_{m=0}^{N-1}$ is given by

$$\hat{\theta}_{c_r^l,c_s}(m) = \hat{c_r}(-(m+l))\hat{c_s}^*(-m), \quad 0 \leq m < N \tag{4}$$

by Lemma 5. To take the IDFT on $\{\hat{\theta}_{c_r^l,c_s}(m)\}_{m=0}^{N-1}$, we first need to compute each of the spectrums $\{\hat{c}_r(m)\}_{m=0}^{N-1}$ and $\{\hat{c}_s(m)\}_{m=0}^{N-1}$, respectively.

Lemma 6 ([13]). *Let* $\{c_r(n)\}_{n=0}^{N-1}$ *and* $\{\hat{c}_r(m)\}_{m=0}^{N-1}$ *be a Zadoff-Chu sequence of period* N *and its spectrum, respectively. Then*

$$\hat{c}_r(m) \ = \ \epsilon_r c_r^*(\langle r^{-1}m\rangle_N), \quad 0 \leq m < N \tag{5}$$

where r^{-1} *is the inverse of* r *in* $\mathbb{Z}_N$ *and the constant* ϵ_r *is given by*

$$\epsilon_r = \begin{cases} \left(\frac{2N}{r}\right)_J W_8\sqrt{N}, & if\ \langle N\rangle_2 = 0\ and\ \langle r\rangle_4 = 1 \\ \left(\frac{2N}{r}\right)_J W_8^{-1}\sqrt{N}, & if\ \langle N\rangle_2 = 0\ and\ \langle r\rangle_4 = 3 \\ \left(\frac{r\alpha}{N}\right)_J c_r(\beta)\sqrt{N}, & if\ \langle N\rangle_4 = 1 \\ \left(\frac{r\alpha}{N}\right)_J W_4\, c_r(\beta)\sqrt{N}, & if\ \langle N\rangle_4 = 3 \end{cases} \tag{6}$$

with $\alpha = \frac{N+1}{2}$ *and* $\beta = \frac{N-1}{2}$*.*

The case that $r = 0$ occurs only when $N = 1$. In this case, ϵ_0 is set to 1 for our convenience. According to Lemma 6, the spectrum of a Zadoff-Chu sequence $\{c_r(\tau)\}$ is a scaled complex-conjugate of the Zadoff-Chu sequence decimated by r^{-1}. Since $\gcd(N, r) = 1$, the decimation by r^{-1} can be simply regarded as a permutation.

Remark 7. *The Jacobi symbols in (6) never vanish, since* $\gcd(r, N) = 1$ *and* $\gcd(\alpha, N) = 1$*. Therefore, we have* $|\epsilon_r| = \sqrt{N}$ *in Lemma 6.*

Using the closed-form expression of the spectrum of a Zadoff-Chu sequence given in Lemma 6, we are now ready to compute $\{\hat{\theta}_{c_r^l,c_s}(m)\}$. Note that $\gcd(N, r) = 1$ and $\gcd(N, s) = 1$ for $N \geq 2$ guarantee the existence of r^{-1} and s^{-1} in $\mathbb{Z}_N$, respectively. Since $g = \gcd(N, r-s)$ and $\langle rr^{-1}\rangle_N = \langle ss^{-1}\rangle_N = 1$, we have $\gcd(N, r^{-1} - s^{-1}) = g$, so $w \triangleq \frac{r^{-1}-s^{-1}}{g}$ is relatively prime to u. Thus, it is possible to construct the Zadoff-Chu sequence $\{c_w(n)\}_{n=0}^{u-1}$ of period u, given by

$$c_w(n) = W_u^{\frac{wn(n+\langle u\rangle_2)}{2}}, \quad 0 \leq n < u. \tag{7}$$

The following lemma gives an important relation between $\{\hat{\theta}_{c_r^l,c_s}(m)\}_{m=0}^{N-1}$ and $\{c_w(n)\}_{n=0}^{u-1}$.

Theorem 8. *Let* $\{c_r(n)\}$ *and* $\{c_s(n)\}$ *be two Zadoff-Chu sequences of period* N*. Let* $g = \gcd(N, r-s)$*,* $u = N/g$ *and* $w = \frac{r^{-1}-s^{-1}}{g}$*. Let* $\{c_w(n)\}_{n=0}^{u-1}$ *be the Zadoff-Chu sequence of period* u *defined as in (7). Then, for* $0 \leq l < N$ *and* $0 \leq m < N$*,*

$$\hat{\theta}_{c_r^l,c_s}(m) = \gamma\, c_w^*(\langle m\rangle_u)W_N^{-\delta m}$$

where $\gamma = \epsilon_r \epsilon_s^* c_r^* (\langle -r^{-1} l \rangle_N)$ *and*

$$\delta = \begin{cases} r^{-1}l - \frac{gw\langle u\rangle_2}{2}, & \textit{if } \langle N\rangle_2 = 0 \\ & \textit{and } \left\langle \frac{rr^{-1}-1}{N} \right\rangle_2 = \left\langle \frac{ss^{-1}-1}{N} \right\rangle_2 \\ r^{-1}l - \frac{N}{2} - \frac{gw\langle u\rangle_2}{2}, & \textit{if } \langle N\rangle_2 = 0 \\ & \textit{and } \left\langle \frac{rr^{-1}-1}{N} \right\rangle_2 \neq \left\langle \frac{ss^{-1}-1}{N} \right\rangle_2 \\ r^{-1}l - \frac{gw(u+1)}{2}, & \textit{if } \langle N\rangle_2 = 1. \end{cases}$$

Proof. Combining (4), (5) and the fact that

$$c_r(n_1+n_2) = c_r(n_1)c_r(n_2)W_N^{rn_1n_2},$$

we get

$$\begin{aligned} \hat{\theta}_{c_r^l, c_s}(m) &= \hat{c}_r(-(m+l))\hat{c}_s^*(-m) \\ &= \epsilon_r \epsilon_s^* c_r^* \left(\langle -r^{-1}l\rangle_N\right) \\ &\quad \cdot c_r^* \left(\langle -r^{-1}m\rangle_N\right) c_s \left(\langle -s^{-1}m\rangle_N\right) W_N^{-r^{-1}lm} \\ &= \epsilon_r \epsilon_s^* c_r^* \left(\langle -r^{-1}l\rangle_N\right) W_N^{-\langle f_{r,s}(m)\rangle_N} W_N^{-r^{-1}lm} \end{aligned}$$

where

$$\begin{aligned} f_{r,s}(m) \triangleq \frac{1}{2} \Big(& r \langle -r^{-1}m\rangle_N \left(\langle -r^{-1}m\rangle_N + \langle N\rangle_2\right) \\ & - s \langle -s^{-1}m\rangle_N \left(\langle -s^{-1}m\rangle_N + \langle N\rangle_2\right) \Big). \end{aligned}$$

It is easily seen that $f_{r,s}(m)$ is an integer for any $0 \le m < N$ and any r, s such that $\gcd(r,N) = \gcd(s,N) = 1$. Here, our problem is divided into two cases based on whether there exists 2^{-1} mod N.

Case 1. $\langle N\rangle_2 = 0$: In this case, we have

$$f_{r,s}(m) = \frac{r\langle -r^{-1}m\rangle_N^2 - s\langle -s^{-1}m\rangle_N^2}{2},$$

but there does not exist 2^{-1} in $\mathbb{Z}_N$. Note that

$$\left\langle r\langle -r^{-1}m\rangle_N^2 - s\langle -s^{-1}m\rangle_N^2 \right\rangle_N = \left\langle \left(r^{-1} - s^{-1}\right) m^2 \right\rangle_N.$$

Using the fact that $W_N^{x/2} = W_N^{y/2}(-1)^q$ for any positive integer x such that $x = y + qN$ with two integers y and q, we get

$$\begin{aligned} W_N^{-\langle f_{r,s}(m)\rangle_N} &= W_N^{-\frac{(r^{-1}-s^{-1})m^2}{2}} (-1)^{\langle q_{r,s}(m)\rangle_2} \\ &= W_u^{-w\frac{m^2}{2}} (-1)^{\langle q_{r,s}(m)\rangle_2} \end{aligned}$$

where

$$q_{r,s}(m) = \frac{\left(r\langle -r^{-1}m\rangle_N^2 - s\langle -s^{-1}m\rangle_N^2\right) - (r^{-1}-s^{-1})m^2}{N}.$$

It is easily checked that

$$\left\langle \frac{\left(r\langle -r^{-1}m\rangle_N^2 - s\langle -s^{-1}m\rangle_N^2\right)}{N} - \frac{(rr^{-1}r^{-1}m^2 - ss^{-1}s^{-1}m^2)}{N} \right\rangle_2 = 0$$

which means

$$\langle q_{r,s}(m)\rangle_2 = \left\langle \frac{(rr^{-1}r^{-1} - ss^{-1}s^{-1})m^2 - (r^{-1}-s^{-1})m^2}{N} \right\rangle_2.$$

Since $rr^{-1} = aN + 1$ and $ss^{-1} = bN + 1$ for some integers a and b, we have

$$\begin{aligned} \langle q_{r,s}(m)\rangle_2 &= \left\langle \frac{\left(r^{-1}aN - s^{-1}bN\right) m^2}{N} \right\rangle_2 \\ &= \left\langle \left(r^{-1}a - s^{-1}b\right) m^2 \right\rangle_2 \\ &= \begin{cases} 0, & \text{if } \langle m\rangle_2 = 0 \\ 0, & \text{if } \langle m\rangle_2 = 1 \text{ and } \langle a\rangle_2 = \langle b\rangle_2 \\ 1, & \text{if } \langle m\rangle_2 = 1 \text{ and } \langle a\rangle_2 \neq \langle b\rangle_2 \end{cases} \end{aligned}$$

for $0 \le m < N$. Hence, we get

$$(-1)^{\langle q_{r,s}(m)\rangle_2} = \begin{cases} 1, & \text{if } \langle a\rangle_2 = \langle b\rangle_2 \\ (-1)^m, & \text{if } \langle a\rangle_2 \neq \langle b\rangle_2 \end{cases}$$

for $0 \le m < N$. When r and s are given, both a and b are fixed. Thus we have

$$\hat{\theta}_{c_r^l, c_s}(m) = \begin{cases} \epsilon_r \epsilon_s^* c_r^* \left(\langle -r^{-1}l\rangle_N\right) \\ \cdot W_u^{-w\frac{m^2}{2}} W_N^{-r^{-1}lm}, & \text{if } \langle a\rangle_2 = \langle b\rangle_2 \\ \epsilon_r \epsilon_s^* c_r^* \left(\langle -r^{-1}l\rangle_N\right) \\ \cdot W_u^{-w\frac{m^2}{2}} W_N^{\left(\frac{N}{2} - r^{-1}l\right)m}, & \text{if } \langle a\rangle_2 \neq \langle b\rangle_2. \end{cases} \quad (8)$$

Note that

$$W_u^{w\frac{m^2}{2}} = \begin{cases} c_w(\langle m\rangle_u), & \text{if } \langle u\rangle_2 = 0 \\ c_w(\langle m\rangle_u) W_u^{-w\frac{m}{2}}, & \text{if } \langle u\rangle_2 = 1 \end{cases} \quad (9)$$

for $0 \le m < u$. Substituting (9) into (8), we get the desired result.

Case 2. $\langle N\rangle_2 = 1$: In this case, we have

$$\begin{aligned} f_{r,s}(m) = \frac{1}{2} \Big(& r\langle -r^{-1}m\rangle_N \left(\langle -r^{-1}m\rangle_N + 1\right) \\ & - s\langle -s^{-1}m\rangle_N \left(\langle -s^{-1}m\rangle_N + 1\right) \Big) \end{aligned}$$

and there exists 2^{-1} in $\mathbb{Z}_N$, i.e, $\langle 2^{-1}\rangle_N = \frac{N+1}{2}$. Note that $W_N^{-\frac{N+1}{2}} = W_N^{\frac{N-1}{2}}$ and

$$\begin{aligned} & \Big\langle r\langle -r^{-1}m\rangle_N \left(\langle -r^{-1}m\rangle_N + 1\right) \\ & \quad - s\langle -s^{-1}m\rangle_N \left(\langle -s^{-1}m\rangle_N + 1\right) \Big\rangle_N \\ & = \left\langle (r^{-1} - s^{-1})m^2 \right\rangle_N. \end{aligned}$$

Hence,

$$\begin{aligned} W_N^{-\langle f_{r,s}(m)\rangle_N} &= W_N^{\frac{N-1}{2}\langle gwm^2\rangle_N} \\ &= W_u^{w\frac{m^2}{2}ug} W_u^{-w\frac{m^2}{2}} \\ &= (-1)^{wm} W_u^{-w\frac{m^2}{2}} \\ &= W_N^{\frac{ug}{2}wm} W_u^{-w\frac{m^2}{2}} \end{aligned}$$

where the third equality comes from the fact that $\langle g\rangle_2 = \langle u\rangle_2 = 1$ when $\langle N\rangle_2 = 1$. Since the Zadoff-Chu sequence $\{c_w(m)\}_{m=0}^{u-1}$ of period u is given by

$$c_w(m) = W_u^{w\frac{m^2+m}{2}}, \quad 0 \le m < u,$$

we have $W_N^{-\langle f_{r,s}(m)\rangle_N} = c_w^*(\langle m\rangle_u) W_N^{\frac{wg(u+1)}{2}m}$. The rest of the proof can be made in a similar way to Case 1. $\square$

Once r and s are given, both w and $\{c_w^*(m)\}_{m=0}^{u-1}$ are directly determined. Theorem 8 tells that $\{\hat{\theta}_{c_r^l,c_s}(m)\}_{m=0}^{N-1}$ can be obtained by g-fold repeating $\{c_w^*(m)\}_{m=0}^{u-1}$, scaling it by γ, and then multiplying it by $W_N^{-\delta m}(-1)^{\langle (N+1)q_{r,s}(m)\rangle_2}$. Note that both γ and δ are constants for a fixed integer $0 \le l < N$.

IV. Closed-Form Expressions of Induced Correlations for Zadoff-Chu Sequences

In the previous section, we derived completely the spectrums $\{\hat{\theta}_{c_r^l,c_s}(m)\}_{m=0}^{N-1}$ of the induced correlations $\{\theta_{c_r^l,c_s}(\tau)\}_{\tau=0}^{N-1}$ in a closed-form. We now get the induced correlation back to the original domain indexed τ by taking the IDFT on the spectrums.

Theorem 9. *Let $\{c_r(n)\}$ and $\{c_s(n)\}$ be two Zadoff-Chu sequences of period N. Let $g = \gcd(N, r-s)$, $u = N/g$, $v = (r-s)/g$ and $w = (r^{-1} - s^{-1})/g$. Then, for $0 \le l < N$ and $0 \le \tau < N$,*

$$\theta_{c_r^l,c_s}(\tau) = \begin{cases} \frac{\gamma \epsilon_w^*}{u} W_u^{w\frac{\tau'(\tau'+\langle u\rangle_2)}{2}}, & \text{if } \langle \tau\rangle_g = \langle -\delta\rangle_g \\ 0, & \text{otherwise} \end{cases}$$

where $\tau' = \frac{w^{-1}}{g}(\tau + \langle \delta\rangle_N)$, and ϵ_w, γ and δ are defined as in Lemma 6 and Theorem 8.

Proof. Let $\{c_w(m)\}_{m=0}^{u-1}$ be the Zadoff-Chu sequence of period u constructed with w. Applying the g-fold repeater $R_g(\cdot)$ in Definition 4 to $\{c_w^*(m)\}$, we define the sequence $\{d(m)\}_{m=0}^{N-1}$ of length N as

$$\{d(m)\}_{m=0}^{N-1} \triangleq R_g\big(\{c_w^*(m)\}_{m=0}^{u-1}\big).$$

Then by Theorem 8, $\hat{\theta}_{c_r^l,c_s}(m) = \gamma d(m) W_N^{-\delta m}$ for $0 \le m < N$. Since γ and δ are constants when l is fixed, we have

$$\begin{aligned}\{\theta_{c_r^l,c_s}(\tau)\}_{\tau=0}^{N-1} &= \mathcal{F}^{-1}\left(\{\hat{\theta}_{c_r^l,c_s}(m)\}_{m=0}^{N-1}\right) \\ &= \gamma \mathcal{F}^{-1}\left(\{d(m)W_N^{-\delta m}\}_{m=0}^{N-1}\right) \\ &= \gamma E_g\big(\mathcal{F}^{-1}(\{c_w^*(m)\}_{m=0}^{u-1})\big)|_{\tau\to\tau+\langle\delta\rangle_N} \quad (10)\end{aligned}$$

where the last equality comes from c) and d) of Lemma 5. It is also easily checked from (1) and (5) that

$$\begin{aligned}\mathcal{F}^{-1}(\{c_w^*(m)\}_{m=0}^{u-1}) &= \left\{\frac{\epsilon_w^*}{u} c_w(\langle w^{-1}\tau\rangle_u)\right\}_{\tau=0}^{u-1} \\ &= \left\{\frac{\epsilon_w^*}{u} W_u^{w\frac{\langle w^{-1}\tau\rangle_u(\langle w^{-1}\tau\rangle_u + \langle u\rangle_2)}{2}}\right\}_{\tau=0}^{u-1}.\end{aligned}$$

Therefore, from (3) and (10), we have

$$\theta_{c_r^l,c_s}(\tau) = \begin{cases} \frac{\gamma\epsilon_w^*}{u} W_u^{w\frac{\frac{w^{-1}}{g}(\tau+\langle\delta\rangle_N)\left(\frac{w^{-1}}{g}(\tau+\langle\delta\rangle_N)+\langle u\rangle_2\right)}{2}}, & \\ & \text{if } \langle\tau\rangle_g = \langle -\delta\rangle_g \\ 0, & \text{otherwise}\end{cases}$$

for $0 \le \tau < N$. $\square$

Theorem 9 tells us that given a pair of Zadoff-Chu sequences $\{c_r(n)\}$ and $\{c_s(n)\}$ of period N, the lth induced correlation $\{\theta_{c_r^l,c_s}(\tau)\}_{\tau=0}^{N-1}$ is simply the g-fold-expanded Zadoff-Chu sequence $\{c_w(\tau)\}_{\tau=0}^{u-1}$ of period u scaled by a constant $\frac{\gamma\epsilon_w^*}{u}$. Since $|\gamma| = N$ and $|\epsilon_w^*| = \sqrt{u}$ by Remark 7, the magnitude of any nonzero component of $\{\theta_{c_r^l,c_s}(\tau)\}$ is equal to $g\sqrt{u}$ which is in agreement with the result in [5].

V. Conclusion

We investigated the induced correlations of Zadoff-Chu sequences in the transform domain. As a result, we presented the induced correlations of two Zadoff-Chu sequences $\{c_r(n)\}$ and $\{c_s(n)\}$ in a closed form. The closed-form expressions indicate that their lth induced cross-correlation can be viewed as a g-fold expanded Zadoff-Chu sequence of period u with constant-scaling. Since our result gives information on the magnitude and phase, it may be applied to further research on partial-period correlations of Zadoff-Chu sequences, including a tighter upper bound on the magnitudes of partial-period correlations than previously known bounds.

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